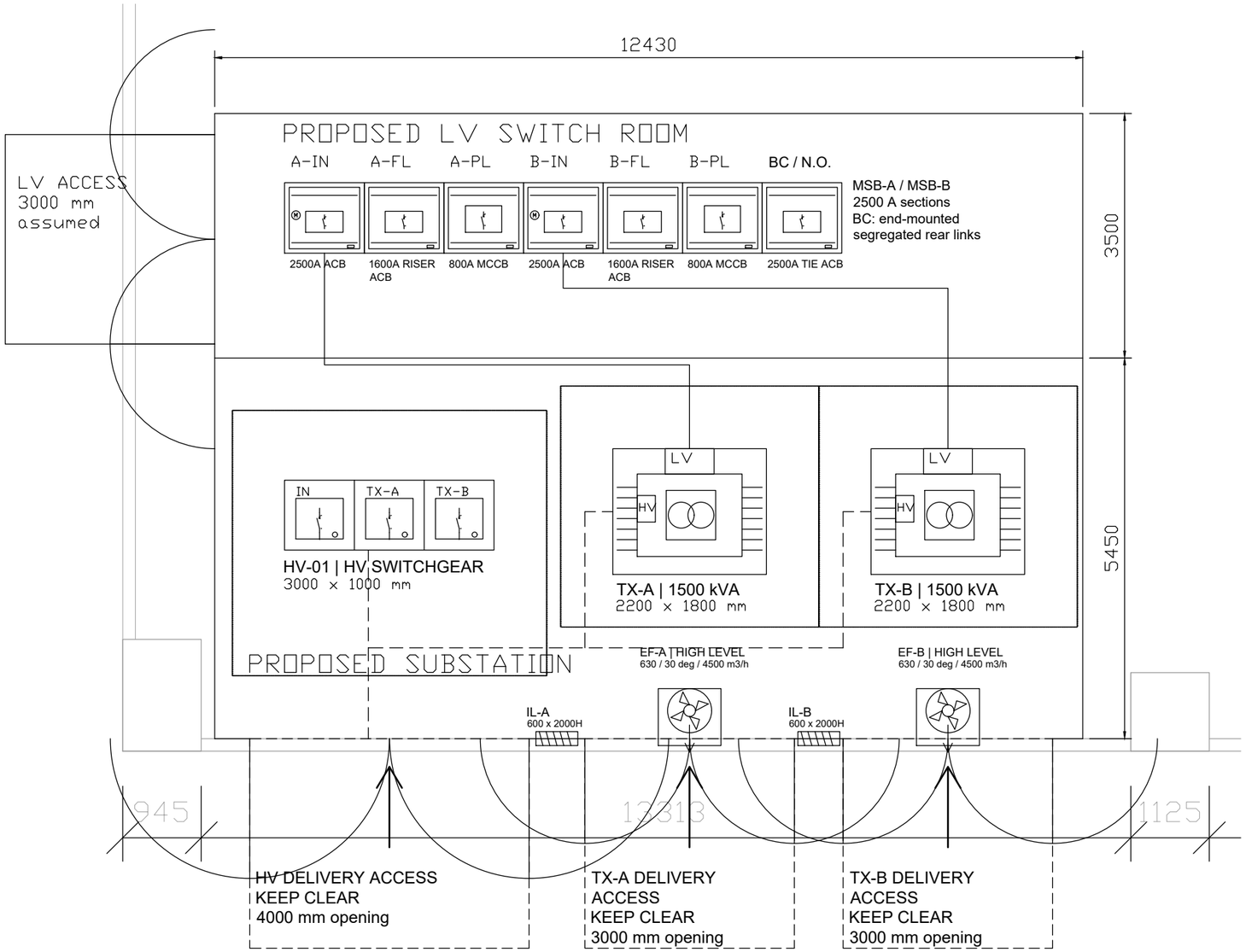
TASK 1 | PROPOSED SUBSTATION AND MAIN LV ROOM

A3 LANDSCAPE | SCALE 1:75 | DIMENSIONS IN mm | PLANTROOM DETAIL



LINE KEY: dashed path = HV trench; solid path = LV route. Dashed boxes = delivery clearance; arrows = delivery direction. Equipment details diagrammatic.

LV INTERFACES / HV ROUTES

LV trench: 1000 W x 1100D, four MCTs per interface. A: 8.03+2=10.03m; B: 8.31+2=10.31m (<20m). 2m is assumed total termination/vertical allowance.

HV centre-lines: 1000 W at panels; 600 W branches; final 300 W; generally 1400D. Segregate HV/LV. Covered trenches; no beam obstruction.

DETAIL 1:75 AT A3

CAPACITY AND EQUIPMENT

TX-A/B: CLP 1500 kVA; 3000 kVA installed, 2672.216 kVA demand. No full-load N-1.

HV-01: incomer + two transformer protection panels.

MSB: seven 1140 mm bays; 2500 A bus sections. BC: 4P normally-open end coupler, segregated insulated rear links to A/B.

A-IN/B-IN/BC electrically and mechanically interlocked. Transfer after isolation and load shedding.

Installation envelopes are design assumptions, not certified manufacturer dimensions.

CLEARANCES / BUILDING WORKS

HV rear1000, front1800, sides750. TX LV terminals north:900; other sides750.

LV board: front1500, rear990; seven bays x1140 =7980 mm.

Proposed FRR partitions; fire-seal penetrations. FFL +0.15 m vs external +0.00 assumed.

Actual structure and equipment dimensions require coordination.

ACCESS AND DOORS

South external access assumed. Provide 3000-wide clear delivery route; unloading distance <=20 m to verify.

Plant doors: HV 4000 W; TX 3000 W each; 2800 H assumed. Outward opening, shown at 90 degrees.

Adjacent plant doors used sequentially; TX-A has wicket personnel access.

Independent LV entrance west.

VENTILATION

EF-A/B: COP101 v16 6.6 type 630 mm / 30 deg / 700 rpm; each 4500 m3/h at 70 Pa, <=58 dB(A) at 1 m.

Design heat: 2 x 15 kW x 1.25 = 37.5 kW. Combined 9000 m3/h gives 12.44 K air rise.

IL-A/B: each 600 W x 2000H, minimum 50% free area; combined 1.20 m2 >=1.12 m2.

High-level ducts: 900 x 750 mm; discharge >=2.5 m above street, away from people. See ventilation calculation.

CLP COP215 Rev07; COP101 v16, 4.3 and 6.1-6.7.

Equipment selections and calculations accompany this drawing.